GROUP - 8

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PAYMENT FAILURE INTELIGENCE

GITHUB.COM/AAYUSHAGASHE/OPUS_ASSIGNMENTS/PAYFAILINTEL

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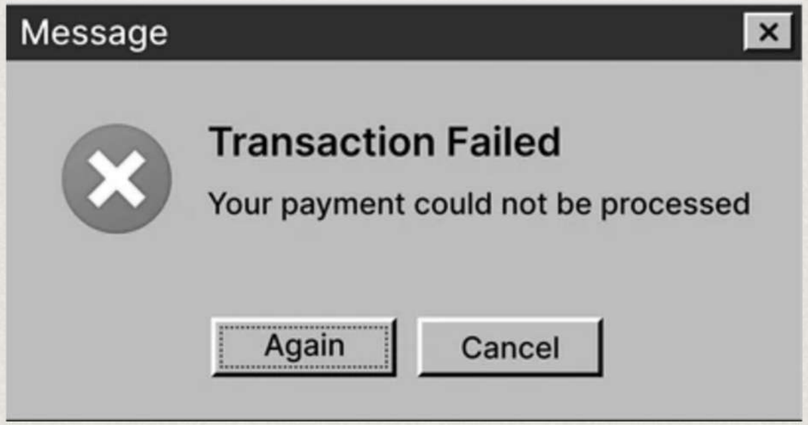
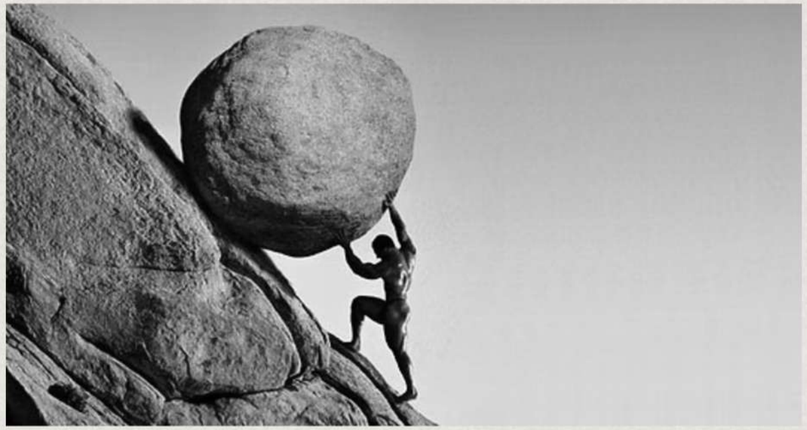




MILLIONS OF DIGITAL PAYMENTS FAIL EVERY DAY ACROSS CARDS, UPI, AND NET-BANKING

FAILURE REASONS ARE CRYPTIC
(E.G. U16, DO NOT HONOR, RTO)

SCATTERED INFORMATION:

- BANK RESPONSES
- PAYMENT GATEWAY LOGS
- INTERNAL TROUBLESHOOTING DOCUMENTS

VSUPPORT TEAMS AND USERS STRUGGLE TO:

- UNDERSTAND FAILURES
- EXPLAIN THEM
- RESOLVE ISSUES QUICKLY

RESULT: INCREASED SUPPORT EFFORT, DELAYED RESOLUTION, POOR USER EXPERIENCE

PROBLEM STATEMENT

ABOUT *the* PROJECT

A GENAL-POWERED PAYMENT FAILURE INTELLIGENCE COPILOT THAT:

UNDERSTANDS PAYMENT FAILURE LOGS AND DOCUMENTS

GENERATES CLEAR, HUMAN-READABLE EXPLANATIONS

RETRIEVES RELEVANT CONTEXT USING RAG

SUGGESTS ACTIONABLE NEXT STEPS

RUNS LOCALLY, WITH STRONG SAFETY CONTROLS

SYSTEM ARCHITECTURE

USER
(Streamlit UI)
- Text Query
- File Upload

STREAMIT APP
(App.py)
- Mode Selection
- Chart Rendering

PRE PROCESSING LAYER
(Preprocessing.py)
- Instruction Removal
- Text Cleaning

DATA INGESTION AYER
(Ingestion.py)
- CSV Reader
- PDF/DOCX/TXT PARSER

CHUNKING LAYER
(Chunking.py)
- Section-aware chunks

EMBEDDING MODEL
(Sentence Transformers)
- Document , Query Embeddings

VECTOR DATABASE
(FAISS)
- Persistent Index
- Temporary Index

RAG ENGINE
(rag_engine.py)
- Error Guardrails
- Hybrid Retrieval
- Prompt Construction

LOCAL LLM
(phi2_llm.py)
- Phi-2 GGUF
- Text Generation

STREAMLIT OUTPUT
- Explaination
- Actions
- Charts

DATA USED

GITHUB

Pubiic-sourced payment dataset

RELATIONAL STRUCTURE:

- *USERS*
- *MERCHANTS*
- *BANKS*
- *TRANSACTIONS (CORE TABLE)*

KEY FIELDS:

- *ERROR CODE*
- *FAILURE REASON*
- *GATEWAY*
- *BANK*

PROCESSED DATA IS EMBEDDED AND STORED AS FAISS ARTIFACTS AS OUR KNOWLEDGE BASE

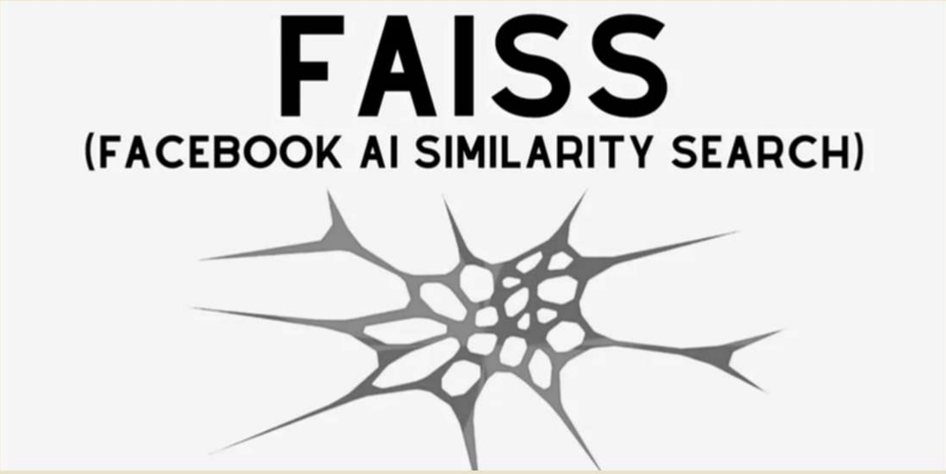
PYTHON - FAST PROTOTYPING AND ECOSYSTEM SUPPORT

SENTENCE TRANSFORMERS - LI HTWEIGHT, LOCAL EMBEDDINGS

FAISS - FAST, LOCAL VECTOR SEARCH

PHI-2 (GGUF) - LOCAL LLM, CPU-OPTIMIZED

TECHNOLOGY STACK



MODEL & AI RESULTS

CORRECTNESS

CORRECT EXPLANATIONS FOR KNOWN ERROR CODES

NO-HAUCINATION

UNKNOWN ERRORS RETURN "INSUFFICIENT INFORMATION"

LATENCY

COLD START 15.99 SECONDS
WARM INFERENCE 0.02S (STEADY-STATE)

DYNAMIC

ANALYTICS ARE QUERY-AWARE (NOT GLOBAL)

CHOICES

HYBRID MODE IMPROVES ANSWER QUALITY

ANALYTICS

SUPPORTS TOP-1/ TOP-3 / TOP-5 FAILURE ANALYSIS
- AUTOMATICALLY ADAPTS IF FEWER REASONS EXIST

HORIZONTAL BAR CHART SHOWS:
- RELATIVE FREQUENCY
- COMPARISON AT A GLANCE

Ask your question:

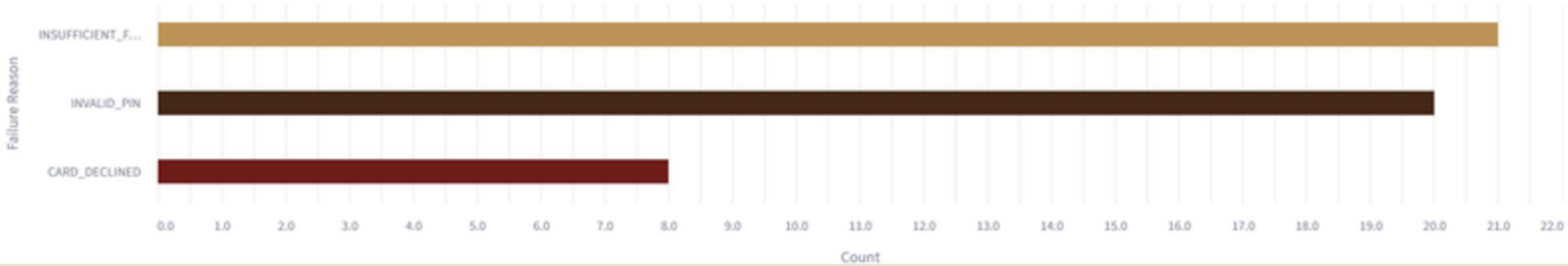
Explain common failures for Axis bank.

Analyze

Top Failure Insights

1. INSUFFICIENT_FUNDS: This occurs when the customer's account doesn't have enough funds to cover the transaction. To reduce this, the bank can implement a system that checks the account balance before processing the transaction.
2. INVALID_PIN: This occurs when the customer enters an incorrect PIN. To reduce this, the bank can implement a system that prompts the customer to enter their PIN multiple times before processing the transaction.
3. CARD_DECLINED: This occurs when the bank's system rejects the card. To reduce this, the bank can implement a system that checks the card's validity before processing the transaction.

Top Failure Reasons



CHALLENGES FACED

- SLOW LLM
- SEMANTIC DRIFT
- HALLUCINATION RISK

KEY LEARNINGS

RAG CORRECTNESS DEPENDS ON RETRIEVAL + GUARDRAILS, NOT MODEL SIZE

SEMANTIC SIMILARITY MUST BE CONSTRAINED FOR SAFETY

ANALYTICS MUST BE DATA-DRIVEN, NOT LLM-DRIVEN

USING QUANTIZED LLM REDUCES LATENCY

FUTURE IMPROVEMENTS

01 *Fine-grained source confidence scoring*

02 *Bank-specific analytics dashboards*

03 *Real-time log ingestion*

04 *Model fine-tuning for payment domain*

05 *Alerting for emerging failure patterns*

THANK YOU

Aayush Agashe